

Table 1: LSDA improvement over baselines for configuration dependency *identification*.

System	cDep	GPTuner	CoT	LLM-RAG
	Mean (improvement)	Mean (improvement)	Mean (improvement)	Mean (improvement)
Precision				
HDFS	0.125 (177.4%)	0.048 (624.8%)	0.097 (258.7%)	0.476 (-27.0%)
HBASE	0.094 (349.7%)	0.029 (1362.4%)	0.059 (618.8%)	0.371 (14.3%)
YARN	0.406 (33.9%)	0.086 (532.2%)	0.111 (389.8%)	0.420 (29.5%)
MAPREDUCE	0.141 (176.2%)	0.062 (526.8%)	0.147 (164.4%)	0.275 (41.3%)
CORE	0.110 (392.5%)	0.036 (1402.2%)	0.073 (640.8%)	0.330 (64.1%)
GLANCE	—	0.075 (528.1%)	0.095 (397.0%)	0.355 (33.3%)
CINDER	—	0.056 (551.2%)	0.037 (877.6%)	0.159 (127.8%)
Average	0.175 (225.9%)	0.056 (789.7%)	0.088 (478.2%)	0.341 (40.5%)
Recall				
HDFS	0.519 (27.9%)	0.429 (54.8%)	0.729 (-8.9%)	0.369 (79.9%)
HBASE	0.455 (90.8%)	0.460 (88.7%)	0.369 (135.2%)	0.598 (45.2%)
YARN	0.608 (54.6%)	0.354 (165.5%)	0.607 (54.9%)	0.472 (99.2%)
MAPREDUCE	0.422 (110.9%)	0.244 (264.8%)	0.530 (67.9%)	0.419 (112.4%)
CORE	0.155 (427.7%)	0.080 (922.5%)	0.553 (47.9%)	0.541 (51.2%)
GLANCE	—	0.064 (1326.6%)	0.155 (489.0%)	0.268 (240.7%)
CINDER	—	0.090 (887.8%)	0.245 (262.9%)	0.218 (307.8%)
Average	0.432 (97.9%)	0.246 (247.6%)	0.455 (87.9%)	0.412 (107.5%)
F1-score				
HDFS	0.202 (125.6%)	0.086 (429.9%)	0.170 (168.1%)	0.139 (227.4%)
HBASE	0.156 (237.4%)	0.055 (858.2%)	0.101 (421.8%)	0.236 (123.4%)
YARN	0.487 (41.3%)	0.138 (398.4%)	0.187 (267.8%)	0.067 (928.1%)
MAPREDUCE	0.211 (155.2%)	0.098 (449.7%)	0.230 (134.2%)	0.067 (702.8%)
CORE	0.129 (402.6%)	0.049 (1219.2%)	0.129 (401.1%)	0.148 (337.3%)
GLANCE	—	0.069 (804.1%)	0.117 (429.5%)	0.126 (391.8%)
CINDER	—	0.069 (648.4%)	0.064 (707.2%)	0.124 (314.4%)
Average	0.237 (140.5%)	0.081 (603.7%)	0.143 (298.6%)	0.284 (100.7%)